12 A retailer sells bags of flour which are advertised as containing 1.5 kg of flour. A trading standards officer is investigating whether there is enough flour in each bag. He collects a random sample and uses software to carry out a hypothesis test at the 5% level. The analysis is shown in the software printout below.

Distribution | Statistics

Z Test of a Mean

Null Hypothesis $\mu =$ 1.5

Alternative Hypothesis ◉ < ○ > ○ ≠

Sample

Mean 1.44

σ 0.24

N 32

Z Test of a Mean

Result

Mean	1.44
σ	0.24
SE	0.0424
N	32
Z	-1.4142
P	0.0786

(a) State the hypotheses the officer uses in the test, defining any parameters used. **[2]**

(b) State the distribution used in the analysis. **[3]**

(c) Carry out the hypothesis test, giving your conclusion in context. **[3]**